

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018 - 2019 (Even Semester)

Innovative Practices Description

UNIT V (4G NETWORKS)

Degree, Semester & Branch: VI Semester B.E. ECE A

Course Code & Title: EC6802 & Wireless Networks

Name of the Faculty member: Mr.P.Gunasekaran

Name of the Topic: OFDM

Name of the Innovative Practice: Animation video

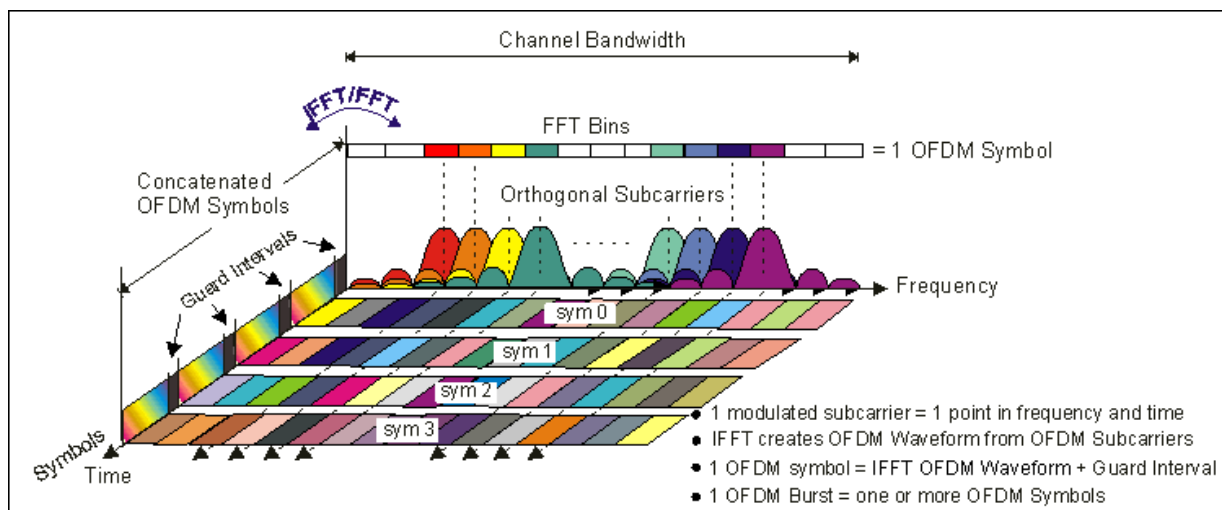
Description:

Orthogonal Frequency Division Multiplexing (OFDM) is a digital multi-carrier modulation scheme that extends the concept of single subcarrier modulation by using multiple subcarriers within the same single channel. Rather than transmit a high-rate stream of data with a single subcarrier, OFDM makes use of a large number of closely spaced orthogonal subcarriers that are transmitted in parallel.

Each subcarrier is modulated with a conventional digital modulation scheme (such as QPSK, 16QAM, etc.) at low symbol rate. However, the combination of many subcarriers enables data rates similar to conventional single-carrier modulation schemes within equivalent bandwidths.

OFDM is based on the well-known technique of Frequency Division Multiplexing (FDM). The OFDM scheme differs from traditional FDM in the following interrelated ways:

1. Multiple carriers (called subcarriers) carry the information stream,
2. The subcarriers are orthogonal to each other, and
3. A guard interval is added to each symbol to minimize the channel delay spread and intersymbol interference.



Frequency-Time Representative of an OFDM signal

References:

1. <https://www.youtube.com/watch?v=tM1lh5KD0GI>
2. http://rfmw.em.keysight.com/wireless/helpfiles/89600b/webhelp/subsystems/wlan-ofdm/Content/ofdm_basicprinciplesoverview.html